

ABSTRACT

ADVANCING FOREX PREDICTION WITH TRANSFORMER-BASED DEEP LEARNING MODELS

Foreign exchange rate forecasting plays a crucial role in predicting currency price movements, optimizing trading strategies, and improving decision-making in the financial industry. The base approach often utilizes simple models to forecast exchange rates based on historical data, but these may not fully capture the complex dependencies in the data. To address this, we enhance forecasting accuracy by employing Gated Recurrent Units (GRU), Informer, and Temporal Fusion Transformers (TFT). These models process data differently to capture both temporal patterns and intricate relationships, providing a comparative analysis of their effectiveness.

The dataset used in this study includes historical exchange rate data spanning 1970 to 2020, encompassing 86 features per timestep, such as lagged exchange rates and additional market-related indicators. Preprocessing steps include handling 8722 missing dates through interpolation, removing 238157 outliers (exchange rates > 10), and feature scaling using Min-Max normalization to standardize data distributions, ensuring optimal model performance.

We implement three advanced deep learning models: GRU, Informer, and TFT. GRU offers a simplified yet effective recurrent structure, Informer leverages attention mechanisms to focus on significant temporal patterns, and TFT combines attention with probabilistic forecasting for improved accuracy. Model performance is evaluated using key metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), Directional Accuracy, Weighted Mean Absolute Percentage Error (WMAPE), Theil's U Statistic, and a custom Profitability Metric that simulates trading outcomes with a 0.1% transaction cost, a 1.0% threshold, and a daily trade limit of 10 trades.

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